PHYS 143 – Mechanics and Thermal Physics

Gravitation

Why do planets, the Moon and satellites move as they do,
and which one force law and one energy explain them all?

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By the end of this week, you should be able to:

1. Apply Kepler's three laws, including the ellipse, the eccentricity and the speeds at perihelion and aphelion.
2. Calculate with Newton's law of gravitation (vector form, superposition), g above and below the surface, and weight against gravity.
3. Analyse circular orbits: v, T and the mass of the central body; geostationary orbits.
4. Use $U = -GMm/r$ for energy conservation, escape speed and the energy of a satellite.

The detailed outcomes are in the checklist at the end of the lecture.

Two contacts:

A (Tuesday): DESCRIBE, then begin to MODEL
How do the planets move, and what force acts?

B (Wednesday): MODEL, then CONSERVE AND APPLY
How do we use the force and the energy for orbits, launches and escape?

Reading: Young and Freedman, Chapter 13, §13.1–13.7.

Labels: **CORE** must do; **PRACTICE** builds the skill; **REFERENCE** keep for later.

Reused, not re-taught

Week 1: vector addition, components.
Weeks 2 and 3: $a = v^2/r$ and $\sum F_{\text{radial}} = mv^2/r$ (circular dynamics, Toolkit 7).
Week 4: work by a varying force.
Week 5: potential energy, reference level, $K + U$ conserved.

Core skills you should master

- use a, e, $r_p = a(1-e)$, $r_a = a(1+e)$;
- compare periods with $T^2 \propto a^3$;
- write $\vec{F}$ with a direction and add forces;
- measure r from the centre;
- set $GMm/r^2 = mv^2/r$ for a circular orbit;
- use $U = -GMm/r$ with $K + U$ conserved.

Supporting tools

The ellipse geometry, the shell theorem, vis-viva and the derivations are in the supplement; slides marked *supporting tool* stay for reference.

IDENTIFY

What is being asked?
What quantities are known?
What physics is relevant?

SET UP

Draw a diagram.
Choose coordinates.
Predict the approximate size and direction.
Select the equations.

EXECUTE

Perform the calculation.
Carry units.
Keep track of signs.

EVALUATE

Are the units correct?
Is the magnitude reasonable?
Does the direction agree with the sketch?
Does it make physical sense?

This week the SET UP step also asks: **where is the centre, and what is r?** Every gravity result depends on a distance measured from a centre.

But what if... we first need observations to explain. Mars is 1.52 times as far from the Sun as Earth: how long is its year, what is the shape of its path, and does it move at a steady speed?

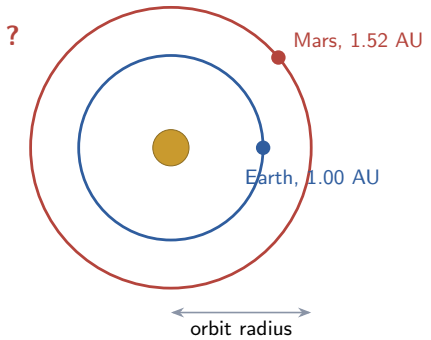
Opening problem

Mars orbits the Sun at $1.52\,\mathrm{AU}$ ($1\,\mathrm{AU}$ is the Earth–Sun distance). Earth takes 1.00 year.

(a) How long is a Martian year?
(b) What is the shape of the path?
(c) Does Mars move at a steady speed?

Sketch first, then predict

Is the Martian year shorter or longer than one Earth year? Is it 1.52 years?



We solve (a) with Kepler's third law, and (b) and (c) with the first two.

But what if... we cannot answer without data. How far, and how long, for each planet?

Contact A: Describe, then begin to Model

How do the planets move, and what force explains it?

Planet	a (AU)	T (years)	e
Mercury	0.386	0.241	0.206
Venus	0.720	0.615	0.007
Earth	1.00	1.00	0.017
Mars	1.52	1.88	0.093
Jupiter	5.19	11.86	0.048
Saturn	9.53	29.45	0.054

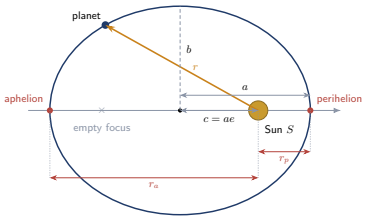
Units and symbols

$1\,\mathrm{AU} = 1.50 \times 10^{11}\,\mathrm{m}$, the mean Earth–Sun distance.
a: the **semi-major axis**, the planet's average distance (defined next).
T: the orbital period.
e: the eccentricity (defined two slides on).

Interpret: farther planets take longer; but is T proportional to a? Mars: 1.52 times as far, 1.88 times as long. No.

But what if... the data give distances, not shapes. If a planet's distance is only *on average* a, the distance must change. What path does that?

Kepler's first law: orbits are ellipses



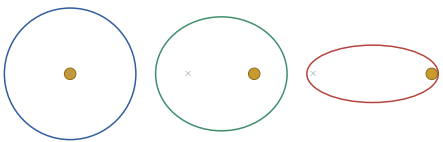
Rule: Kepler's first law

Each planet moves in an **ellipse** with the **Sun at one focus**.

Definitions. a: semi-major axis, half the longest dimension. c: centre to a focus. **Perihelion:** the closest point, r_p. **Aphelion:** the farthest point, r_a. Nothing is at the other focus.

But what if... the Sun is at a focus, not the centre. How lopsided is the ellipse? We need one number for the shape.

Eccentricity: how elongated is the ellipse?



Rule: eccentricity

$e = \dfrac{c}{a}, \quad 0 \le e < 1.$

$e = 0$ is a circle;

$e \to 1$ is a very long thin ellipse.

Rule: closest and farthest

$r_p = a(1 - e)$,

$r_a = a(1 + e)$

so $r_p + r_a = 2a$.

The planets in the table have $e < 0.21$, almost circles (Venus 0.007, Earth 0.017, Mercury 0.206).

But what if... comets do not have almost circular orbits. How do we find r_p and r_a in numbers?

- **TOOLKIT 1 – Orbit geometry**

WHEN TO USE: a problem gives a, e, or the nearest and farthest distances of an elliptical orbit.

PROCEDURE: **1. SKETCH** the ellipse, the Sun at a focus, and mark perihelion and aphelion.
2. LIST two of a, e, r_p, r_a; mark the Sun's side.
3. USE $r_p = a(1-e)$, $r_a = a(1+e)$, $r_p + r_a = 2a$.
4. CONVERT to a and e if the other two are given.

WATCH OUT: distances are measured *from the Sun* (the focus), not from the centre of the ellipse; a is not r_p.

CHECK: $r_p < a < r_a$ and $0 \leq e < 1$; $r_p + r_a = 2a$.

Example 1: Comet Halley

Comet Halley is $8.80 \times 10^7\,\mathrm{km}$ from the Sun at perihelion and $5.25 \times 10^9\,\mathrm{km}$ at aphelion. Find its semi-major axis and eccentricity.

Identify: a and e, from r_p and r_a.

Set up: sketch a long ellipse with the Sun near one end; $2a = r_p + r_a$ and $r_p = a(1-e)$. Predict: $r_p \ll r_a$, so e is close to 1.

Execute:

$$a = \frac{r_p + r_a}{2} = \frac{8.80 \times 10^7 + 5.25 \times 10^9}{2} = 2.67 \times 10^9\,\mathrm{km}$$

$$e = 1 - \frac{r_p}{a} = 1 - \frac{8.80 \times 10^7}{2.67 \times 10^9} = 0.967$$

Evaluate: e close to 1 ✓. In AU: $a = 17.8\,\mathrm{AU}$, $r_p = 0.587\,\mathrm{AU}$ (inside Venus's orbit) and $r_a = 35.0\,\mathrm{AU}$ (beyond Neptune).

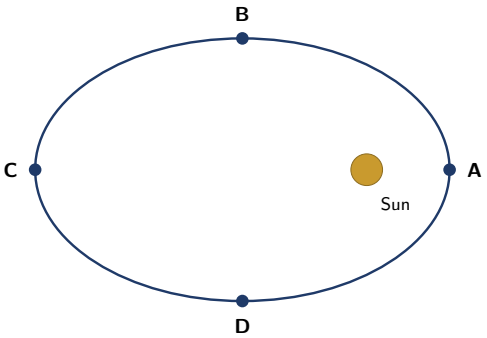
Your turn

For Earth, $a = 1.50 \times 10^{11}\,\mathrm{m}$ and $e = 0.017$. Predict whether r_p and r_a differ greatly, then calculate them.

Answer

$1.47 \times 10^{11}\,\mathrm{m}$ and $1.53 \times 10^{11}\,\mathrm{m}$: a 3.4% difference.

But what if... the Sun's distance changes along the orbit. Does the planet move at the same speed all the way round?



Sketch first, then predict

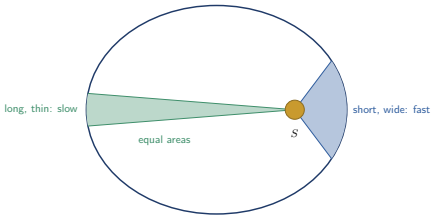
The planet passes A, B, C and D (A is perihelion, C is aphelion).

Where is it moving fastest? Slowest? Does the planet's speed stay constant?

Prediction guide: nearer the Sun, the Sun pulls harder and the planet speeds up, like a ball rolling down into a valley. We now state this exactly.

But what if... *faster* and *slower* are not enough. Is there a rule that tells us exactly how the speed changes along the orbit?

Kepler's second law: equal areas in equal times



Rule: Kepler's second law

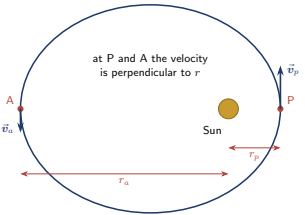
A line from the Sun to the planet sweeps out **equal areas in equal times**:

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \text{constant}$$

The two shaded regions are swept in the *same* time. Near the Sun the region is short and wide, so the planet covers a long arc: it moves **fast**. Far from the Sun it covers a short arc: **slow**.

The rate of sweeping is $\dfrac{\mathrm{d}A}{\mathrm{d}t} = \frac{1}{2} r v \sin\phi = \dfrac{L}{2m}$, with L the planet's angular momentum about the Sun. So equal areas means L is constant (Lecture 8).

But what if... this tells us the speed varies, but not by how much. At the two ends of the orbit, can we turn *equal areas* into numbers?



Rule: speeds at the two ends

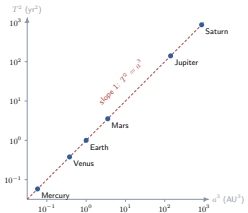
$v_p\, r_p = v_a\, r_a$

so $\dfrac{v_p}{v_a} = \dfrac{r_a}{r_p}$: the planet is faster at perihelion.

Why: at P and A the velocity is perpendicular to r, so in a short time Δt the swept area is a thin triangle, $\frac{1}{2} r\, v\, \Delta t$. Equal areas give $\frac{1}{2} r_p v_p \Delta t = \frac{1}{2} r_a v_a \Delta t$.

Energy view: $K + U$ is constant and $U = -GMm/r$ is lowest when r is smallest, so K and v are greatest at perihelion and smallest at aphelion.

But what if... Kepler 1 and 2 describe *one* planet (the tutorial gives real speeds). What links the planets to each other, so that Mars's year follows from its distance?



Rule: Kepler's third law

The square of the period is proportional to the cube of the semi-major axis:

$$T^2 \propto a^3$$

Interpret: on log axes T^2 against a^3 is a straight line of slope 1. In years and AU, $T^2/a^3 \approx 1.0$ for every planet in the table. So $T \propto a^{3/2}$, and a is the *semi-major axis*, not the whole major axis.

The constant is the same for all bodies orbiting the *same* central body.

But what if... we do not always have a table. Given *one* orbit, how do we get another quickly, without any constants?

- **TOOLKIT 2 – Ratios and scaling**

WHEN TO USE: a quantity follows a power law ($T \propto a^{3/2}$, later $F \propto 1/r^2$) and a known case is given.

PROCEDURE: **1. WRITE** the law as a ratio: $\dfrac{T_2}{T_1} = \left(\dfrac{a_2}{a_1}\right)^{3/2}$.

2. PREDICT bigger or smaller before calculating.

3. SUBSTITUTE the ratio of the *same* quantity in the same units; constants cancel.

WATCH OUT: use the ratio of the semi-major axes, not of the periods, in the bracket; the exponent is $3/2$, not $2/3$.

CHECK: units of the ratio cancel; a bigger orbit has the longer period.

Example 2: the Martian year (the opening problem) CORE

Identify: T_{Mars}, given $a_{\text{Mars}} = 1.52\,\text{AU}$ and Earth ($a = 1\,\text{AU}$, $T = 1\,\text{yr}$).
Set up: same Sun, so Toolkit 2. Predict: longer than 1 year, but *not* 1.52 years (the law has the power $3/2$).
Execute:

$$T = 1.00\,\text{yr} \times \left(\frac{1.52}{1.00}\right)^{3/2} = 1.87\,\text{yr}$$

Evaluate: $1.87 > 1.52$, as predicted; $1.87 \times 365 \approx 685$ days (measured: 687 days). **The opening problem is answered:** (a) about 1.9 years; (b) an ellipse with $e = 0.093$ (Kepler 1); (c) no, faster at perihelion (Kepler 2).

Your turn

Europa and Io orbit Jupiter. Io: $a = 4.217 \times 10^5\,\text{km}$, $T = 1.769$ days. Europa: $a = 6.709 \times 10^5\,\text{km}$. Predict, then find Europa's period.

Answer

$T = 1.769\,(6.709/4.217)^{3/2} = 3.55$ days. (Scaling $T \propto a$ would wrongly give 2.81 days.)

But what if... the law works for Jupiter's moons too, but with a *different* constant than for the planets. Why does the constant depend on the central body?

Decide for each statement, then compare with your neighbour

I. The line from the Sun to a planet sweeps equal areas in equal times.

II. A planet moves faster at aphelion than at perihelion.

III. $T^2 \propto a^3$, where a is the semi-major axis.

IV. The Sun is at the centre of a planet's elliptical orbit.

Answers

I and III are correct.
II is false: the planet is *faster* at perihelion ($v_p r_p = v_a r_a$ with $r_p < r_a$).
IV is false: the Sun is at a focus.

But what if... the answer to II came from $v_p r_p = v_a r_a$. Is there a reason in the forces for this? We have only described motion so far.

Mistake	Correction
Putting the Sun at the centre of the ellipse.	The Sun is at a *focus*; $r_p \neq r_a$.
Using $r_p = a(1-e)$ with a as the *major axis*.	a is the *semi*-major axis, half the length.
Writing $T \propto a^2$ or $T \propto a^{2/3}$.	$T^2 \propto a^3$, so $T \propto a^{3/2}$.
Saying a planet is slowest at perihelion.	Fastest at perihelion, slowest at aphelion.
Thinking Earth's seasons come from its distance from the Sun.	They come from the tilt of the axis.
Using T in years but a in metres in a ratio.	Keep the same units for the same quantity in a ratio.

But what if... Kepler found *how* the planets move, with no reason why. What force, acting at a distance, makes $T^2 \propto a^3$?

A new question: from describing to explaining

So far

Kepler's laws **describe** orbits:
- ellipses, with $r_p = a(1-e)$ and $r_a = a(1+e)$;
- equal areas in equal times;
- $T^2 \propto a^3$.

They say nothing about *why*, or about satellites and falling objects.

We now ask

What **force** acts between the Sun and a planet, or the Earth and a satellite?
How does it change with distance and with the masses?
How do we calculate it for several bodies, and what does it mean for a falling object?

The rest of Contact A in one line

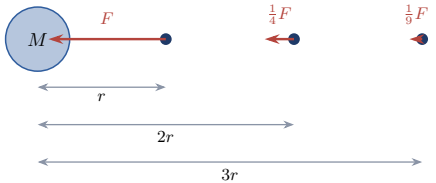
physical idea (a force from Kepler's third law) $\rightarrow$ model (Newton's law, vector form) $\rightarrow$ method (superposition, the field g, weight on another planet).

Take a planet of mass m in a **circular** orbit of radius r (the $e \approx 0$ case), with $T^2 = k\,r^3$ (k constant):

$$v = \frac{2\pi r}{T} \quad \Rightarrow \quad a_{\text{rad}} = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$$
$$= \frac{4\pi^2 r}{k\,r^3} = \frac{4\pi^2}{k}\,\frac{1}{r^2}$$

Newton's second law (Week 3, circular dynamics) gives the net force toward the Sun:

$$F = ma_{\text{rad}} = \frac{4\pi^2 m}{k}\,\frac{1}{r^2}$$



Rule: what Kepler 3 implies

The force on a planet falls as $\frac{1}{r^2}$: double the distance, one quarter the force.

But what if... $4\pi^2/k$ is a constant of the Sun, and the force is proportional to m. What is that constant, and what about the *other* body's mass?

Rule: Newton's law of gravitation

Every particle attracts every other particle with a force directed along the line joining them, whose magnitude is

$$F = \frac{G m_1 m_2}{r^2}, \qquad G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2/kg^2}$$

Interpret

- proportional to *both* masses ($m_1 m_2$);
- inverse square of the distance r between the *centres* (spheres act as particles at their centres).

Newton's third law

The two forces are an action–reaction pair: equal in size, opposite in direction, acting on *different* bodies. The Earth pulls a $70.0\,\mathrm{kg}$ person with $686\,\mathrm{N}$; the person pulls the Earth with $686\,\mathrm{N}$.

But what if... the law is for particles. How large is the force between two everyday objects, and does the larger one feel the larger force?

Example 3: two small spheres

Two spheres, $m_1 = 0.0100\,\mathrm{kg}$ and $m_2 = 0.500\,\mathrm{kg}$, have centres $0.0500\,\mathrm{m}$ apart, far from other objects. Find (a) the gravitational force on each, (b) the acceleration of each.

Identify: F on each sphere, then a_1 and a_2.

Set up: spheres act as particles at their centres. Third law: the two forces are equal. Predict: tiny forces, and the lighter sphere accelerates more.

Execute:

$$F = \frac{Gm_1m_2}{r^2} = \frac{(6.67\times10^{-11})(0.0100)(0.500)}{(0.0500)^2}$$

$$= 1.33\times10^{-10}\,\mathrm{N}$$

$$a_1 = \frac{F}{m_1} = 1.33\times10^{-8}\,\mathrm{m/s^2}$$

$$a_2 = \frac{F}{m_2} = 2.66\times10^{-10}\,\mathrm{m/s^2}$$

Evaluate: equal forces, unequal accelerations ($a_1 = 50\,a_2$); not constant, as the force grows when they approach.

Misconception

"The larger body pulls harder." The forces are equal in size. The larger body accelerates less because its mass is larger.

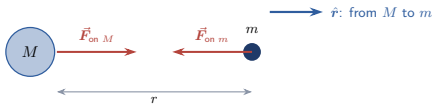
Your turn

The centres are moved to $0.100\,\mathrm{m}$. Predict the new force, then find it.

Answer

One quarter: $3.33\times10^{-11}\,\mathrm{N}$ (twice the distance, $1/2^2$ of the force).

But what if... with one other body the direction is obvious. With several bodies, how do the pulls combine?



Rule: vector form

The force on m due to M:

$$\vec{F}_{\text{on } m} = -\frac{GMm}{r^2}\,\hat{r}$$

$\hat{r}$ is the unit vector pointing **from** M **to** m; the minus sign makes the force point back toward M.

Check: $|\vec{F}| = GMm/r^2$; direction toward the other body. Swapping roles reverses $\hat{r}$, so $\vec{F}_{\text{on } M} = -\vec{F}_{\text{on } m}$.

But what if... with one other body the direction is obvious. With several bodies, how do the pulls combine?

- **TOOLKIT 3 – Superposition of gravitational forces**

WHEN TO USE: a mass is attracted by two or more others (a triangle of masses, the Earth–Moon–Sun).

PROCEDURE: **1. SKETCH** the masses and draw one arrow *toward each other body* on the mass of interest.
2. PREDICT the direction of the net force (between the arrows) and which arrow is longer.
3. CALCULATE each magnitude $F_i = Gm\,M_i/r_i^2$ with its *own* distance r_i.
4. ADD as vectors: components (Week 1, Toolkit 5), then magnitude and direction.

WATCH OUT: do not add the magnitudes unless the forces are along one line; the forces act on the same body, so there is no action–reaction cancelling.

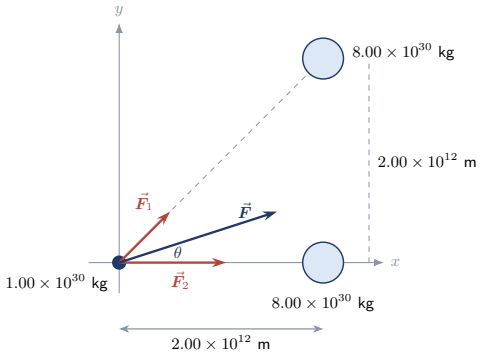
CHECK: the net force points into the sketched region; it is smaller than the sum of the magnitudes.

Many stars belong to systems held together by their mutual gravitational attraction. In a three-star system the stars sit at the vertices of a $45°$ right triangle with legs $2.00 \times 10^{12}\,\mathrm{m}$. The two large stars each have mass $8.00 \times 10^{30}\,\mathrm{kg}$; the small star has mass $1.00 \times 10^{30}\,\mathrm{kg}$.

Find the total gravitational force on the small star.

Sketch first, then predict

Which large star pulls harder? Will the net force point closer to the x-axis or to the line at $45°$?



Example 4: three stars, solution

CORE

Identify: total force on the small star: superposition (Toolkit 3).

Set up: $\vec{F}_1$ from the star at the end of the diagonal, $\vec{F}_2$ from the star along $+x$; use components; $L = 2.00 \times 10^{12}\,\mathrm{m}$, $GMm = 5.34 \times 10^{50}\,\mathrm{N\,m^2}$. Predict: $F_2 > F_1$ (nearer), so the net is closer to $+x$.

Execute:

$$F_1 = \frac{GMm}{2L^2} = 6.67 \times 10^{25}\,\mathrm{N}$$

$$F_2 = \frac{GMm}{L^2} = 1.33 \times 10^{26}\,\mathrm{N}$$

$$F_x = F_1 \cos 45^\circ + F_2 = 1.81 \times 10^{26}\,\mathrm{N}$$

$$F_y = F_1 \sin 45^\circ = 4.72 \times 10^{25}\,\mathrm{N}$$

$$F = \sqrt{F_x^2 + F_y^2} = 1.87 \times 10^{26}\,\mathrm{N}$$

$$\theta = \tan^{-1}\frac{F_y}{F_x} = 14.6^\circ$$

Evaluate: $14.6^\circ < 45^\circ$, close to $+x$ ✓; $F < F_1 + F_2$ ✓. A huge force, yet $a = F/m = 1.87 \times 10^{-4}\,\mathrm{m/s^2}$.

Your turn

A $1.0\,\mathrm{kg}$ mass m is at the origin, $M_1 = 4.0\,\mathrm{kg}$ at $(2.0\,\mathrm{m}, 0)$ and $M_2 = 6.0\,\mathrm{kg}$ at $(0, 3.0\,\mathrm{m})$. Predict the quadrant of the net force, then find it.

Answer

$F_1 = 6.67 \times 10^{-11}\,\mathrm{N}$,
$F_2 = 4.45 \times 10^{-11}\,\mathrm{N}$;
$F = 8.02 \times 10^{-11}\,\mathrm{N}$ at 33.7° above $+x$ (first quadrant). With M_2 at $(0, -3.0\,\mathrm{m})$: the same size, 33.7° *below* $+x$.

But what if... gravity decides the motion of stars, but a ball near the Earth feels $9.8\,\mathrm{N/kg}$. When one mass is the whole Earth, what is the simplest way to describe the pull?

Rule: gravitational acceleration at distance r from a spherical mass M

$$g(r) = \frac{GM}{r^2} \qquad \text{toward the centre}$$

At the Earth's surface $g_s = GM_E/R_E^2 \approx 9.80\,\mathrm{m/s^2}$; we use $g_s = 9.80\,\mathrm{m/s^2}$ throughout.

Why the mass of the object does not appear

$F = mg = GMm/r^2$: m cancels, so all objects fall with the same acceleration at one place (Week 2). Near the surface $g \approx 9.80\,\mathrm{m/s^2}$ is constant while $h \ll R_E$.

Weighing the Earth

$m_E = \dfrac{g_s R_E^2}{G} = \dfrac{(9.80)(6.37\times10^6)^2}{6.67\times10^{-11}} = 5.96\times10^{24}\,\mathrm{kg}$.

Mean density: $m_E/(\frac{4}{3}\pi R_E^3) = 5500\,\mathrm{kg/m^3}$.

Interpret: surface rocks have a density of only 2000 to $3300\,\mathrm{kg/m^3}$, so the interior must be denser than the average: the Earth is not uniform (the centre is about $13\,000\,\mathrm{kg/m^3}$).

But what if... Is the same law useful on another planet? What does a lander weigh on Mars?

Example 5: gravity on Mars

A robotic lander has an Earth weight of $3430\,\mathrm{N}$. It is sent to Mars ($R_M = 3.39 \times 10^6\,\mathrm{m}$, $m_M = 6.42 \times 10^{23}\,\mathrm{kg}$). Find its weight on Mars and g_M there.

Identify: weight F_g on Mars and g_M.

Set up: find the lander's mass from $w = mg$; then $F_g = Gm_M m/R_M^2$. Predict: smaller than on Earth (Mars has 11% of the mass but 53% of the radius).

Execute:

$$m = \frac{w}{g} = \frac{3430}{9.80} = 350\,\mathrm{kg}$$

$$F_g = \frac{(6.67 \times 10^{-11})(6.42 \times 10^{23})(350)}{(3.39 \times 10^6)^2} = 1.30 \times 10^3\,\mathrm{N}$$

$$g_M = \frac{F_g}{m} = 3.7\,\mathrm{m/s^2}$$

Evaluate: about 38% of the Earth value ✓. The mass, $350\,\mathrm{kg}$, is the same everywhere; only the weight changes.

Your turn

Check g_M directly from $g_M = Gm_M/R_M^2$.

Answer

$\dfrac{(6.67 \times 10^{-11})(6.42 \times 10^{23})}{(3.39 \times 10^6)^2} = 3.7\,\mathrm{m/s^2}$ ✓.

Compare planets

$g \propto M/R^2$. Planet with $M = 4M_E$ and $R = 2R_E$: $4/2^2 = 1$, the same g as on Earth.

Common mistakes: forces and weight

CORE

Mistake	Correction
Adding forces on a mass as numbers.	Add as vectors, each toward its own source (Toolkit 3).
Saying the larger body feels the larger force.	The two forces are equal (third law); the accelerations differ.
Measuring r between the surfaces.	r is between the *centres*.
Writing $F \propto 1/r$ or $1/r^3$.	$F \propto 1/r^2$: doubling r gives one quarter.
Saying the mass of the lander changes on Mars.	The mass is the same everywhere; the weight changes.

Contact A: what we can now do

- use a, e, r_p, r_a and the ratio $T_2/T_1 = (a_2/a_1)^{3/2}$;
- compare speeds at perihelion and aphelion;
- find the force between two masses, and the net force from several;
- find g and the weight on any planet.

Contact B asks

How does g change above and below the surface?
Is weight the same as the gravitational force?
What speed and period does a circular orbit need, and how much energy does a launch take?

But what if... g is a constant only near the ground, and every distance must be measured from the centre. How far can we trust $g = 9.80\,\mathrm{m/s^2}$? $\rightarrow$ Contact B.

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Contact B: Model, then Conserve and apply

How do we use the force and the energy for orbits, launches and escape?

From Contact A

- Kepler 1 to 3: ellipse, equal areas, $T^2 \propto a^3$;
- $F = Gm_1m_2/r^2$, attractive, r between centres;
- forces from several bodies add as vectors;
- $g = GM/r^2$, and weight on another planet.

Check before we start (2 min)

A planet has mass $M = 2M_E$ and radius $R = R_E$. Predict whether g at its surface is $2g_s$, $4g_s$ or $g_s/2$.

Answer

$g \propto M/R^2$, so $g = 2g_s$.

Contact B in one line

$g(r)$ above and below the surface $\rightarrow$ circular orbits (v, T, M) $\rightarrow$ energy, $U = -GMm/r$ $\rightarrow$ escape, satellites and problems where the method is not given.

- **TOOLKIT 4 – Measure r from the centre**

WHEN TO USE: any gravity problem that gives a height, an altitude or a depth.

PROCEDURE: **1. SKETCH** the body, mark its *centre*, its radius R, and the point of interest.

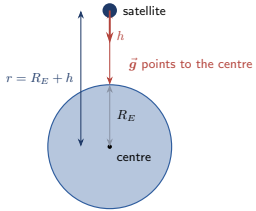
2. FIND r: above the surface $r = R + h$; below the surface $r = R - d$.

3. CHOOSE the formula by region: $g = GM/r^2$ for $r \geq R$; $g = g_s\, r/R$ for $r \leq R$ (next slides).

4. SANITY-CHECK with limits: $h = 0$ gives g_s; $r \to \infty$ gives 0; the centre gives 0.

WATCH OUT: an *altitude* of $400\,\mathrm{km}$ is not $r = 400\,\mathrm{km}$; also never use the surface g_s at altitude.

CHECK: g at altitude is smaller than g_s, and still not zero.



Rule: g at altitude h

$$r = R_E + h, \qquad g = g_s \left(\frac{R_E}{R_E + h} \right)^2$$

Altitude h	r/R_E	g (m/s^2)
surface, 0	1.00	9.80
400 km **(ISS)**	1.06	8.68
$R_E = 6370$ km	2.00	2.45
35 800 km **(geostationary)**	6.62	0.224

An airliner at 10 km: $(6370/6380)^2 = 0.9969$, so you weigh 0.3% less.

But what if... the ISS astronauts float, yet $g = 8.68\,\mathrm{m/s^2}$ there. Is gravity absent in orbit? Let us calculate one case, then explain the floating.

Example 6: gravity at the ISS

The International Space Station orbits at an altitude of $400\,\mathrm{km}$. Find g there.

Identify: g at $h = 4.00 \times 10^5\,\mathrm{m}$.

Set up: Toolkit 4: $r = R_E + h = 6.37 \times 10^6 + 0.400 \times 10^6 = 6.77 \times 10^6\,\mathrm{m}$. Predict slightly less than 9.80: 6% farther from the centre, so about 12% smaller.

Execute:

$$g = 9.80\left(\frac{6.37}{6.77}\right)^2 = 9.80\,(0.885) = 8.68\,\mathrm{m/s^2}$$

Evaluate: $8.68 < 9.80$ ✓(88.5% of g_s).

Misconception

"There is no gravity in orbit." There is: the ISS feels 88.5% of its surface weight. Astronauts float because they are *in free fall together* with the station (their apparent weight is zero).

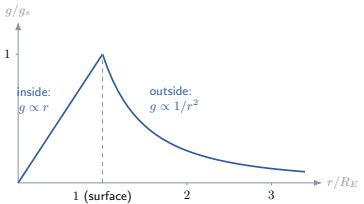
Your turn

At what altitude is $g = g_s/4$?

Answer

$r = 2R_E$, so $h = R_E = 6370\,\mathrm{km}$.

But what if... this formula needs $r \geq R_E$. Dig a mine shaft: does g rise toward infinity as $r \to 0$? Surely not at the centre.



Rule: shell theorem (statement)

A spherical shell of mass exerts **no net force** on a body inside it. At radius $r < R$ only the mass *inside* r pulls. For uniform density, $M(r) = M\,(r/R)^3$, so

$$g(r) = \frac{G\,M(r)}{r^2} = g_s\,\frac{r}{R} \quad (r \le R)$$

Interpret: g grows linearly from 0 at the centre to g_s at the surface, then falls as $1/r^2$. At a depth of $1000\,\mathrm{km}$, $g = 9.80\,(5370/6370) = 8.26\,\mathrm{m/s^2}$. The real Earth has a denser core, so the uniform model gives the shape only. The proof is in the supplement.

But what if... let us use this: what is the force on a parcel dropped down a shaft through the centre?

Example 7: a parcel dropped down a shaft

CORE

Imagine a shaft drilled through the Earth along a diameter, and a parcel of mass m dropped into it. Assume the Earth has uniform density. Find the gravitational force on the parcel as a function of its distance r from the centre.

Identify: $F_g(r)$ for $r < R_E$.

Set up: only the mass M inside radius r pulls (shell theorem). Mass is proportional to volume, $\frac{4}{3}\pi r^3$ against $\frac{4}{3}\pi R_E^3$. Predict: zero at the centre, GM_Em/R_E^2 at the surface.

Execute:

$$M = m_E \frac{r^3}{R_E^3}, \qquad F_g = \frac{GMm}{r^2} = \frac{Gm_Em}{R_E^3}\, r$$

Evaluate: $F_g \propto r$, not $1/r^2$. At $r = R_E$ it equals Gm_Em/R_E^2 ✓; at $r = 0$ it is 0 ✓.

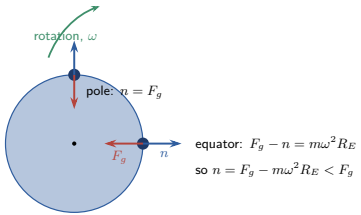
Your turn

At what distance from the centre is g equal to half of g_s?

Answer

$g = g_s r/R_E = g_s/2$ gives $r = R_E/2 = 3185\,\text{km}$.

But what if... we now have the *force* at every place. A scale measures something, though. Is what a scale reads always equal to the gravitational force?



Rule: gravity and weight

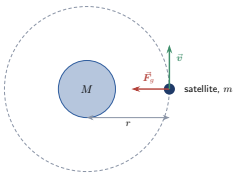
Gravitational force: $F_g = GMm/r^2$, the pull of the Earth.

Weight (apparent weight): what a scale reads, the normal force n on the body.

At rest in an inertial frame, $n = F_g = mg$. Otherwise $n \neq F_g$.

Equator: the person moves in a circle, so $F_g - n = m\omega^2 R_E = mv^2/R_E$. A point on the equator travels $2\pi R_E$ in $86\,164\,\mathrm{s}$: $v = 465\,\mathrm{m/s}$ and $v^2/R_E = 0.0339\,\mathrm{m/s^2}$. So g is about $0.03\,\mathrm{m/s^2}$ (0.35%) smaller at the equator than at the poles, where $n = F_g$. **Free fall** (a lift with a cut cable, an orbit): $n = 0$, *weightless*, although $F_g \neq 0$.

But what if... in orbit $n = 0$, yet $F_g \neq 0$: gravity alone is the inward force, so the satellite never needs an engine to keep going round. What speed and period does that require?



Rule: circular orbit

Gravity is the **only** force and supplies the inward net force:

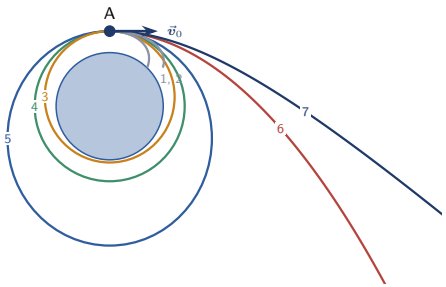
$$\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{GM}{r}}$$

$$T = \frac{2\pi r}{v} = 2\pi\sqrt{\frac{r^3}{GM}}, \quad \omega = \sqrt{\frac{GM}{r^3}}$$

Interpret: $\vec{F}_g$ *is* the net inward force (no extra arrow). Larger orbits mean slower speeds and longer periods. The ISS ($r \approx 6800\,\text{km}$) has $v = 7.7\,\text{km/s}$ and $T = 93$ minutes; the Moon ($r = 384\,000\,\text{km}$) has $v = 1.0\,\text{km/s}$ and $T = 27.3$ days.

But what if... m cancelled: heavy and light satellites at one radius move alike. A circle is only one orbit: what if the launch speed changes?

Orbits of every shape



A projectile is launched from A in the direction tangent to the Earth's surface; trajectories 1 to 7 show the effect of a larger and larger launch speed v_0 (air ignored).

- **1, 2:** not fast enough; it returns to the ground.
- **3, 4, 5:** closed orbits; every closed orbit is an ellipse, and 4 is a circle.
- **6, 7:** open orbits, a parabola and a hyperbola; it never returns.

Apparent weightlessness occurs whenever gravity is the only force, whatever the shape of the orbit.

- **TOOLKIT 5 – The circular-orbit method**

WHEN TO USE: a satellite, moon or planet moves in a circle (or a nearly circular orbit) under gravity alone.

PROCEDURE: **1. SKETCH** the orbit, the centre, $\vec{F}_g$ toward the centre, $\vec{v}$ tangent; mark r from the centre (Toolkit 4).

2. WRITE $\dfrac{GMm}{r^2} = \dfrac{mv^2}{r}$ (or $m\omega^2 r$); cancel m.

3. SOLVE for the unknown: $v = \sqrt{GM/r}$, or r from T, or M from v and r.

4. CONVERT with $v = 2\pi r/T$ to link v and T.

WATCH OUT: the satellite's mass cancels; r is $R + h$, not h; T in seconds.

CHECK: $v \leq 7.91\,\mathrm{km/s}$ for any Earth orbit; lower orbits are faster.

Example 8: the International Space Station

The ISS is in a circular orbit $400\,\mathrm{km}$ above the Earth. Find its speed and period. ($GM_E = 3.98 \times 10^{14}\,\mathrm{m^3/s^2}$.)

Identify: v and T for a circular orbit.

Set up: $r = 6.77 \times 10^6\,\mathrm{m}$ (Toolkit 4). Predict just below the $7.91\,\mathrm{km/s}$ of a surface skimming orbit, and a period near 90 minutes.

Execute:

$$v = \sqrt{\frac{3.98 \times 10^{14}}{6.77 \times 10^6}} = 7.67 \times 10^3\,\mathrm{m/s}$$

$$T = \frac{2\pi r}{v} = \frac{2\pi(6.77 \times 10^6)}{7.67 \times 10^3} = 5.55 \times 10^3\,\mathrm{s} = 92.4\,\mathrm{min}$$

Evaluate: $7.67 < 7.91\,\mathrm{km/s}$ ✓. A small Ghanaian satellite, GhanaSat-1, built by students of All Nations University, was released from the station in 2017.

But what if... this used Newton's law and the radial equation. Does it reproduce Kepler's third law, which we only observed?

Your turn (exam style)

A satellite is $200\,\mathrm{km}$ above the Earth. Find v for a circular orbit. Check: its radial acceleration is $g = GM/r^2$.

Answer

$r = 6.57 \times 10^6\,\mathrm{m}$; $v = 7.79\,\mathrm{km/s}$; $a_{\mathrm{rad}} = v^2/r = 9.23\,\mathrm{m/s^2} = GM/r^2$ ✓.

Rule: Kepler's third law with its constant

For a circular orbit about a mass M,

$$T^2 = \frac{4\pi^2}{GM} r^3$$

and for an ellipse the same with $r \to a$ (derivation in the supplement). For the Sun, $T = 2\pi a^{3/2}/\sqrt{GM_\odot}$, whatever the eccentricity.

Interpret:
- $T^2 \propto a^3$ ✓, as Kepler found;
- the constant $4\pi^2/GM$ depends on the *central* body only: Sun $2.97 \times 10^{-19}\,\mathrm{s^2/m^3}$, Earth $9.91 \times 10^{-14}\,\mathrm{s^2/m^3}$;
- the orbiting body's mass does not appear.

Concept check (exam style)

A satellite of mass m has period T in a circular orbit of radius R about the Earth. If it had mass $4m$ (same R), its period would be
(A) $4T$ (B) $2T$ (C) T (D) $T/2$ (E) $T/4$

Answer: (C)

$T = 2\pi\sqrt{R^3/GM_E}$: m cancels.

But what if... the constant contains M. So measuring an orbit weighs the central body, with no scales at all.

Example 9: the asteroid Pallas

The asteroid Pallas has an orbital period of 4.62 years and an eccentricity of 0.233. Find the semi-major axis of its orbit.

Identify: a, given T and the central body, the Sun ($M_\odot = 1.99 \times 10^{30}\,\text{kg}$).

Set up: Kepler's third law with its constant, $a^3 = GM_\odot T^2/4\pi^2$. Predict: between Mars and Jupiter. The eccentricity is not needed.

Execute: $T = 4.62\,(3.156 \times 10^7\,\text{s}) = 1.46 \times 10^8\,\text{s}$

$$a = \left(\frac{GM_\odot T^2}{4\pi^2}\right)^{1/3}$$

$$= \left(\frac{(6.67 \times 10^{-11})(1.99 \times 10^{30})(1.46 \times 10^8)^2}{4\pi^2}\right)^{1/3}$$

$$= 4.15 \times 10^{11}\,\text{m}$$

Evaluate: $4.15 \times 10^{11}\,\text{m} = 2.77\,\text{AU}$, between Mars ($1.52\,\text{AU}$) and Jupiter ($5.19\,\text{AU}$): the asteroid belt ✓.

Your turn

Find the period of Comet Halley, with $a = 2.67 \times 10^{12}\,\text{m}$ (Example 1). Predict first: more or less than 50 years?

Answer

$T = \dfrac{2\pi a^{3/2}}{\sqrt{GM_\odot}} = 2.38 \times 10^9\,\text{s} \approx 75$ years. It was at perihelion in 1986 and returns in 2061. (Check: $17.8^{3/2} = 75$.)

But what if... these are descriptions of motion. To put a satellite in orbit, someone had to raise it from the ground. How much energy does that take? → energy.

Common mistakes: orbits and g

Mistake	Correction
Using $r = h$ (altitude) instead of $r = R + h$.	r is measured from the *centre*.
Using $g = 9.80\,\mathrm{m/s^2}$ at orbital altitude.	g falls as $1/r^2$: $8.68\,\mathrm{m/s^2}$ at $400\,\mathrm{km}$.
Saying astronauts float because there is no gravity.	Gravity is still $\sim 90\%$ of g_s; they are in free fall.
Adding a "centripetal" force to the diagram.	$\vec{\boldsymbol{F}}_g$ *is* the inward net force.
Thinking a heavier satellite has a different period.	m cancels in $v = \sqrt{GM/r}$ and in T.
Using $g \propto 1/r^2$ inside the Earth.	Inside, $g \propto r$ (uniform model).

So far

Contact A: Kepler's laws describe orbits; $F = GMm/r^2$ explains them.
Contact B so far: $g(r)$ and circular orbits (v, T, M).
All use **forces**, and the force changes with distance.

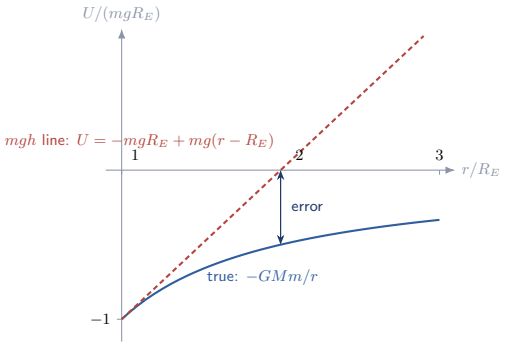
We now ask

How much **energy** does a launch, a climb or an escape need when the force is not constant?
What is the energy of a satellite, and what does its sign tell us?
Can we solve problems where the method is not given?

The rest of Contact B in one line

energy (Week 5) $\rightarrow U = -GMm/r \rightarrow$ escape, satellites and bound orbits $\rightarrow$ choosing the right method for new problems.

Why mgh is not enough



Sketch first, then predict

The energy to lift $1\,\mathrm{kg}$ from the surface to height h: is $g_s h$ exact, too large, or too small, for large h?

h	$g_s h$	true $\Delta U/m$
$400\,\mathrm{km}$	$3.92\,\mathrm{MJ}$	$3.69\,\mathrm{MJ}$
R_E	$62.4\,\mathrm{MJ}$	$31.3\,\mathrm{MJ}$

g falls with height, so mgh *overestimates*: 6% at $400\,\mathrm{km}$ and a factor 2 at $h = R_E$.

But what if... we need the work of a force that varies as $1/r^2$. What function $U(r)$ has $F = -\mathrm{d}U/\mathrm{d}r$ equal to the gravitational force?

Rule: gravitational potential energy

Two masses M and m, centres a distance r apart, have

$$U(r) = -\frac{GMm}{r}, \qquad U \to 0 \text{ as } r \to \infty$$

The reference level ($U = 0$) is at infinite separation; U is always **negative**, rising toward zero as r increases.

Meaning of the minus sign

Gravity pulls masses together, so work must be done *against* it to separate them to infinity: a bound system has *less* energy than the separated one. Working in the supplement.

Link to mgh (Week 5)

$\Delta U = GMm\dfrac{h}{R(R+h)} \approx mg_s h$ for $h \ll R$. At 400 km: 3.69 against 3.92 MJ/kg (6% off).

But what if... how do we use a negative U to find a speed or a height? → a method.

- **TOOLKIT 6 – The energy method for gravity**

WHEN TO USE: speeds, heights or energies at two positions, when r changes by a large amount or the body leaves the Earth.

PROCEDURE: **1. SKETCH** the two states; mark the centre and both distances r_1 and r_2 (Toolkit 4).

2. REFERENCE: $U = 0$ at infinity, so $U = -GMm/r$ (not mgh).

3. WRITE $\frac{1}{2}mv_1^2 - \dfrac{GMm}{r_1} = \frac{1}{2}mv_2^2 - \dfrac{GMm}{r_2}$ (no other forces do work).

4. SOLVE for the unknown; m cancels.

WATCH OUT: U is negative: signs of the terms; r measured from the centre; air and engines are not included.

CHECK: $K \geq 0$ at every position; for $h \ll R$ the answer agrees with mgh.

Example 10: a capsule fired from a giant cannon

CORE

A giant cannon, sunk into the ground, fires a capsule straight up. Neglect air, rotation and the Moon. Find the minimum muzzle speed to reach (a) a height R_E above the surface, (b) infinity (escape).

Identify: v_1 for two end points.

Set up: only gravity does work: $K_1 + U_1 = K_2 + U_2$, $U = -GMm/r$ (Toolkit 6). Point 2 has $v_2 = 0$: $r_2 = 2R_E$ in (a), $r_2 = \infty$ in (b).

Execute:

$$\text{(a)}\quad v_1 = \sqrt{\frac{GM}{R_E}} = 7.91 \times 10^3\,\text{m/s}$$

$$\text{(b)}\quad v_1 = \sqrt{\frac{2GM}{R_E}} = 1.12 \times 10^4\,\text{m/s}$$

Evaluate: neither answer depends on the capsule's mass or the launch direction. A site at latitude 28° already moves east at $410\,\text{m/s}$ (Earth's rotation), so launching east helps.

Your turn

Use (b) to find the escape speed from Mars ($m_M = 6.42 \times 10^{23}\,\text{kg}$, $R_M = 3.39 \times 10^6\,\text{m}$).

Answer

$v = \sqrt{2GM/R} = 5.03 \times 10^3\,\text{m/s}$.

But what if... (b) is a general result: the least speed to leave any planet. Before generalising, one more use of the method: what if the speed is less than escape?

Example 11: how high does a probe go?

PRACTICE

A probe is launched straight up at $8.0\,\mathrm{km/s}$ from the Earth's surface. Ignoring air and rotation, find the greatest height reached.

Identify: $h_{\max}$, where $v = 0$.

Set up: Toolkit 6. Predict: more than the $v^2/2g_s = 3.3 \times 10^6\,\mathrm{m}$ that constant-g would give, because g weakens.

Execute:

$$\tfrac{1}{2}v_0^2 - \frac{GM}{R_E} = -\frac{GM}{r_{\max}}$$

$$\frac{1}{r_{\max}} = 1.570 \times 10^{-7} - 8.04 \times 10^{-8} = 7.66 \times 10^{-8}$$

$$r_{\max} = 1.31 \times 10^7\,\mathrm{m},\ h = 6.68 \times 10^6\,\mathrm{m}$$

Evaluate: $6.68 \times 10^3\,\mathrm{km} > 3.3 \times 10^3\,\mathrm{km}$ ✓; $v_0 < 11.2\,\mathrm{km/s}$, so the probe returns.

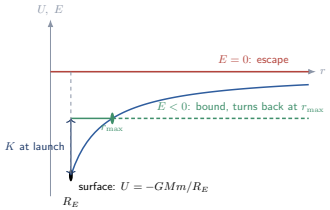
Your turn

Launched at $v_0 = 11.0\,\mathrm{km/s}$, will the probe go higher or lower than $r = 2R_E$? Predict, then show.

Answer

Higher: $1/r_{\max} = 1.570 \times 10^{-7} - 1.519 \times 10^{-7}$, so $r_{\max} = 2.0 \times 10^8\,\mathrm{m} \approx 31R_E$, far beyond $2R_E$.

But what if... at $11.0\,\mathrm{km/s}$ the probe goes $31R_E$ out; a little faster, does it come back at all?



Rule: escape speed

The least launch speed to reach $r \to \infty$ (arriving with $v \to 0$) has $E = 0$:

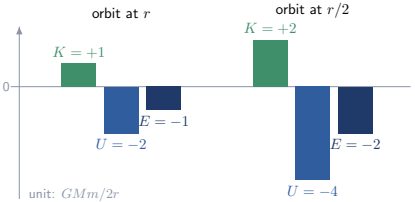
$$\tfrac{1}{2}mv_{\text{esc}}^2 - \frac{GMm}{R} = 0$$

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}} = \sqrt{2}\, v_{\text{circ}}$$

$E < 0$ turns back at $r_{\max}$ (Example 11). m does not appear; M and R do. Earth: $\sqrt{2(3.98 \times 10^{14})/6.37 \times 10^6} = 11.2\,\text{km/s}$.

Body	Moon	Mars	Earth	Jup.	Sun
v_{esc} **(km/s)**	2.37	5.03	11.2	60.2	618

But what if... escape has $E = 0$; a satellite in orbit has $E \neq 0$. What is its energy and sign?



Rule: circular orbit energies

From $mv^2 = GMm/r$:

$$K = \frac{GMm}{2r}, \quad U = -\frac{GMm}{r}$$

$$E = K + U = -\frac{GMm}{2r} = -K = \tfrac{1}{2}U$$

Interpret: $E < 0$ means **bound**. A smaller orbit has *more negative* E and *larger* K. The larger the orbit, the greater (less negative) E and U, and the smaller K.

But what if... air drag removes energy from a satellite. Does it slow down? E gets more negative: what happens to r and to v?

Example 12: a satellite in a $340\,\mathrm{km}$ orbit

A $1000\,\mathrm{kg}$ satellite is to enter a circular orbit $340\,\mathrm{km}$ above the Earth. (a) Find its speed, period and radial acceleration. (b) How much work puts it in orbit? (c) How much more makes it escape?

Identify: v, T, a_{rad} in (a); energy differences in (b) and (c).

Set up: $r = 6370 + 340 = 6710\,\mathrm{km}$ (Toolkit 4). (a) circular-orbit method; (b) and (c) from $E = -GMm/2r$, with $E = -GMm/R_E$ on the ground (at rest) and $E = 0$ at escape. Predict: v near $7.7\,\mathrm{km/s}$, T near 90 min.

Execute (a):

$$v = \sqrt{\frac{GM_E}{r}} = \sqrt{\frac{(6.67\times10^{-11})(5.97\times10^{24})}{6.71\times10^{6}}}$$
$$= 7700\,\mathrm{m/s}$$
$$T = \frac{2\pi r}{v} = \frac{2\pi(6.71\times10^{6})}{7700}$$
$$= 5470\,\mathrm{s} = 91.2\,\mathrm{min}$$
$$a_{\text{rad}} = \frac{v^2}{r} = \frac{(7700)^2}{6.71\times10^{6}} = 8.84\,\mathrm{m/s^2}$$

Evaluate (a)

a_{rad} is the value of g at $340\,\mathrm{km}$, about 10% less than at the surface ✓.

Next slide

The work needed is the difference between the final and the initial total energy.

Example 12: a satellite in a $340\,\mathrm{km}$ orbit, (b) and (c)

Execute (b): on the ground $K_1 = 0$ (ignore rotation), so

$$E_1 = -\frac{GM_E m}{R_E} = -6.25 \times 10^{10}\,\mathrm{J}$$

$$E_2 = -\frac{GM_E m}{2r} = -2.97 \times 10^{10}\,\mathrm{J}$$

$$W = E_2 - E_1 = 3.28 \times 10^{10}\,\mathrm{J}$$

Execute (c): escape needs $E = 0$ ($E = 0$ is the least energy to reach infinity), so

$$W' = 0 - E_2 = 2.97 \times 10^{10}\,\mathrm{J}$$

Evaluate: $E_2 > E_1$ (the orbit has more energy than the ground). The ground-state kinetic energy from the Earth's rotation was ignored; for a $1000\,\mathrm{kg}$ satellite at the equator it is $\frac{1}{2}(1000)(465)^2 = 1.1 \times 10^8\,\mathrm{J}$, about 0.3% of W.

Key idea

The larger the orbit, the greater (less negative) the total energy, the smaller the speed. The radius is always measured from the centre of the Earth.

Sketch first, then predict

A low satellite loses a little energy to the thin atmosphere. After a while, is its speed larger, smaller, or the same? Does its orbit rise or fall?

Misconception

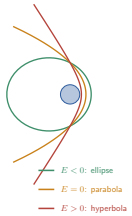
"Drag slows a satellite down." Over the long run it makes the satellite *faster* and lower, until it burns up. To *raise* an orbit, an engine adds energy, and the satellite ends up *slower*.

Reasoning with $E = -GMm/2r$

E decreases $\Rightarrow$ r decreases (orbit falls). Then $v = \sqrt{GM/r}$ *increases*: $K = -E$ rises, because U falls twice as fast as E. Drag *speeds the satellite up* as it spirals in.

But what if... these are circles. Launch it with *less* or *more* speed than the circular value: is the path still closed?

Bound and unbound orbits



Rule: the sign of E fixes the orbit

$E < 0$: closed orbit (ellipse, circle); for an ellipse $E = -\dfrac{GMm}{2a}$ (semi-major axis; derivation and vis-viva in the supplement).
$E = 0$: open (parabola): just escapes.
$E > 0$: open (hyperbola): leaves with speed to spare.

Interpret: a comet with $E < 0$, like Halley's, returns; a one-time visitor from outside has $E \geq 0$. The shapes are the three conic sections; Kepler's laws describe the closed case.

But what if... ratios, forces, energy: which tool for a new problem? → Toolkit 7.

• TOOLKIT 7 – Choose the method

PROCEDURE: 1. IDENTIFY what is asked: a force, a time, a speed, an energy, a mass, a distance.
2. MATCH to the tool, then **check its conditions**:

Clue in the problem	Method	Conditions
two orbits about the same body, a ratio of periods or sizes	Kepler ratio (Toolkit 2)	same central mass
net force, g at a place, forces from several bodies	Newton's law (Toolkits 3, 4)	r from centres
v, T or M for a circular orbit	circular-orbit method (Toolkit 5)	gravity only, circle
speeds or heights at two places, escape, energy to move	energy (Toolkit 6)	$U = -GMm/r$
speed at a point of an ellipse; e, r_p, r_a	Toolkit 1, $v_p r_p = v_a r_a$ (vis-viva: supplement)	closed orbit

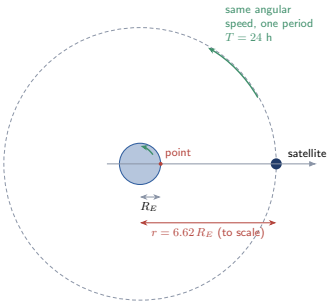
3. CROSS-CHECK with a second tool when two apply (e.g. $E = -GMm/2r$ with $v = \sqrt{GM/r}$).
CHECK: units, sign of E, and the limits $7.9\,\mathrm{km/s}$ and $11.2\,\mathrm{km/s}$.

Application problem: decide what physics applies

A communications satellite stays directly above one point on the equator, for example above the Gulf of Guinea, south of Tema.

(a) At what altitude is it? (b) What is its speed? (c) Why can this happen only above the equator?

Data: the Earth turns once in $86\,164\,\mathrm{s}$;
$GM_E = 3.98 \times 10^{14}\,\mathrm{m^3/s^2}$; $R_E = 6.37 \times 10^{6}\,\mathrm{m}$.



Sketch first, then predict

Is the orbit higher or lower than the Moon's? Will its speed be more or less than the ISS's $7.67\,\mathrm{km/s}$? Which tool does the clue "the period is known" point to?

Example 13: solution

Identify: r and v for a circular orbit with $T = 86\,164\,\text{s}$ (Toolkit 7: circular orbit).
Set up: the satellite must have the Earth's angular speed, so T equals the Earth's period. Gravity alone, circular.
Execute:

$$r = \left(\frac{GM_E T^2}{4\pi^2}\right)^{1/3} = 4.21 \times 10^7\,\text{m}$$

$$h = r - R_E = 3.58 \times 10^7\,\text{m}, \qquad v = \frac{2\pi r}{T} = 3.07 \times 10^3\,\text{m/s}$$

Evaluate: $h = 35\,800\,\text{km}$, well below the Moon ($3.84 \times 10^5\,\text{km}$); $v = 3.07 < 7.67\,\text{km/s}$ (higher orbits are slower) ✓. Check: $v = \sqrt{GM/r} = 3.07\,\text{km/s}$ ✓.

(c) The equator

Gravity points to the Earth's centre, so every orbit lies in a plane through the centre. Only the equatorial plane contains a circle that keeps pace with a point on the equator; any other orbit would swing north and south.

Cross-check with a ratio

The Moon has $T = 27.3$ d at 3.84×10^8 m:
$r = 3.84 \times 10^8\,(1/27.3)^{2/3} = 4.2 \times 10^7$ m ✓(Toolkit 2).

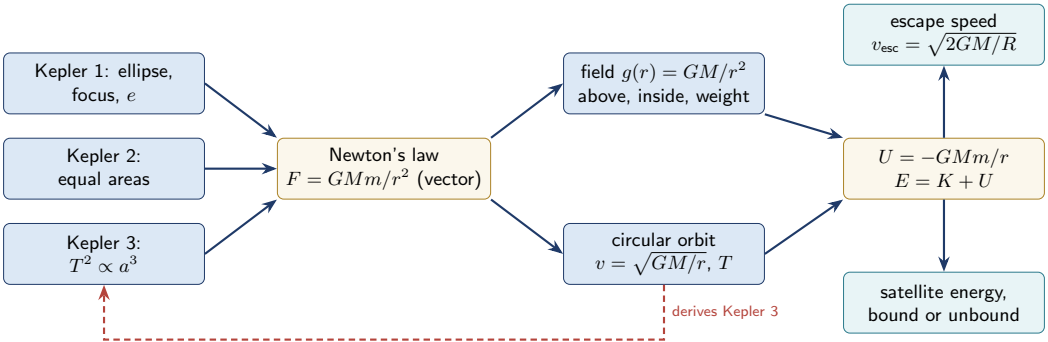
Common mistakes: energy

Mistake	Correction
Using $U = mgh$ for a launch from the Earth to a large height.	$U = -GMm/r$; mgh only for $h \ll R$.
Dropping the minus sign of U, or treating $E < 0$ as impossible.	$E < 0$ means bound; $K \geq 0$ is the real condition.
Using $r =$ altitude in U or v_{esc}.	r from the *centre*: $R + h$.
Thinking the escape speed depends on the mass of the projectile.	m cancels: $\sqrt{2GM/R}$ depends on the planet only.
Believing drag slows a satellite down for good.	E falls, the orbit shrinks, v *rises*.
Using $E = -GMm/2r$ for an elliptical orbit.	$E = -GMm/2a$, with the *semi-major axis*.
Mixing Kepler ratios of different central bodies.	The ratio T^2/a^3 is the same only for one central body.

But what if... we have three tools (ratios, forces, energy). How do they fit together, and where do we use them next? → Summary.

Week 6 concept map

CORE



Key ideas

- **Kepler's laws describe orbits:** ellipse with the Sun at a focus; equal areas in equal times; $T^2 \propto a^3$.
- **Newton's law explains them:** $F = GMm/r^2$, attractive, with r between centres; $\vec{\boldsymbol{F}} = -(GMm/r^2)\,\hat{\boldsymbol{r}}$; forces from several bodies add as vectors.
- $g = GM/r^2$ **outside,** $g = g_s r/R$ **inside (uniform model);** weight is the scale reading, not always F_g; $g_M = GM/R^2$ gives the surface gravity of any planet.
- **Circular orbit:** $v = \sqrt{GM/r}$, $T = 2\pi\sqrt{r^3/GM}$; the satellite's mass cancels; $T^2 = 4\pi^2 r^3/GM$ weighs the central body.
- $U = -GMm/r$ (zero at infinity); $E = K + U$ is conserved; $v_{\text{esc}} = \sqrt{2GM/R}$; $E = -GMm/2r$ (circle), $-GMm/2a$ (ellipse); $E < 0$ means bound.
- Habit: sketch with the centre and r, predict, choose the tool, evaluate.

Reference tools: where to look

Equations, procedures: the next slides (core equations, master procedure, seven toolkits).

Definitions: the equation-anatomy slides.

Common mistakes: the mistakes slides in each contact.

What you should remember: the chain, Contact A

CORE

Concept	Limitation that motivates the next	Leads to
Framework, opening problem	need data: distances and periods of the planets	**Solar-system data**
Solar-system data	T is not proportional to a; the distance is only an average	**Kepler 1**
Kepler 1, eccentricity, Example 1	the Sun's distance changes: is the speed constant?	**Where is it fastest?**
Equal areas (Kepler 2)	faster or slower is not a number: ends of the orbit	$v_p r_p = v_a r_a$
Speeds at the ends	one planet only: what links planets?	**Kepler 3**
Kepler 3, Example 2	describes, but does not explain: what force?	**Newton's law**
Newton's law, Example 3	one source only: several bodies?	**Vector form, Toolkit 3**
Superposition, Example 4	$10^{-11}\,\mathrm{N}$, yet a ball feels $9.8\,\mathrm{N/kg}$	**Field g**
$g = GM/r^2$, Example 5	constant only near the ground	**Contact B**

What you should remember: the chain, Contact B

CORE

Concept	Limitation that motivates the next	Leads to
Toolkit 4, Example 6	needs $r \geq R_E$; what below the surface?	g **inside**
Shell theorem, Example 7	real Earth: what does a scale read?	**Weight against gravity**
Weight	a satellite needs an inward force only from gravity	**Circular orbits**
Toolkit 5, Example 8	does it reproduce Kepler 3?	**Kepler 3 from Newton**
Kepler 3, Example 9	raising a satellite: mgh fails	**Energy**
mgh **fails**	which U gives the inverse-square force?	$U = -GMm/r$
Toolkit 6, Example 10	does the probe ever not come back?	**Escape speed**
Escape speed	what is a satellite's energy?	**Satellite energy**
$E = -GMm/2r$**, Example 12**	drag removes energy: slower or faster?	**Drag paradox**
E **fixes the orbit**	many tools: which one?	**Toolkit 7**
Toolkit 7, Example 13	ratios, forces, energy: how they fit	**Summary**

Reference tools: core equations

Idea	Equation	Conditions and units
Ellipse	$r_p = a(1-e),\ r_a = a(1+e),\ r_p + r_a = 2a$	r from the Sun; a semi-major
Kepler 2 (ends)	$v_p r_p = v_a r_a$	velocity $\perp r$ at both ends
Kepler 3	$T^2 \propto a^3;\ T_2/T_1 = (a_2/a_1)^{3/2};\ T = 2\pi a^{3/2}/\sqrt{GM}$	same central body
Gravitation	$F = Gm_1m_2/r^2;\ \vec{\boldsymbol{F}} = -(GMm/r^2)\hat{\boldsymbol{r}}$	r between centres; N
Field	$g = GM/r^2\ (r \geq R);\ g = g_s r/R\ (r \leq R)$	uniform density inside; m/s^2
Weight	$n = F_g - m\omega^2 R_E$ at the equator	scale reading, N
Mass of a planet	$M = gR^2/G$	g at the surface
Circular orbit	$v = \sqrt{GM/r},\ T = 2\pi\sqrt{r^3/GM},\ T^2 = 4\pi^2 r^3/GM$	gravity only; r from centre
Potential energy	$U = -GMm/r$	zero at infinity; J
Escape speed	$v_{\text{esc}} = \sqrt{2GM/R}$	$E = 0$; no air
Energies	$E = -GMm/2r$ (circle), $-GMm/2a$ (ellipse)	$E < 0$: bound
Vis-viva	$v^2 = GM(2/r - 1/a)$	closed orbit; m/s

• MASTER TOOLKIT – For a gravitation problem

1. IDENTIFY What is asked: force, g, period, speed, energy, mass?
2. SKETCH Body, orbit, the forces or the two states.
3. CENTRE AND r Mark the centre; r from the centre ($R + h$, or $R - d$).
4. PREDICT Bigger or smaller than g_s, 7.9, $11.2\,\mathrm{km/s}$?
5. MODEL Kepler ratio, Newton's law, circular orbit, or energy (Toolkit 7).
6. EXECUTE Algebra first, numbers last; carry units.
7. EVALUATE Units, sign of E, magnitude, limits ($h = 0$, $r \to \infty$).

If the result disagrees with the sketch or the prediction, investigate before accepting it.

Inside the course framework

IDENTIFY (step 1): what is being asked?
SET UP (steps 2 to 5): sketch, centre and r, predict, model.
EXECUTE (step 6): calculate.
EVALUATE (step 7): check the result.

THE SEVEN TOOLKITS

1	**Orbit geometry**	a, e, $r_p = a(1-e)$, $r_a = a(1+e)$, $r_p + r_a = 2a$; Sun at a focus
2	**Ratios and scaling**	$T_2/T_1 = (a_2/a_1)^{3/2}$; constants cancel; predict first
3	**Superposition of forces**	arrow toward each source; magnitudes; add as vectors
4	**Measure r from the centre**	$r = R + h$ or $R - d$; g outside and inside; limit checks
5	**Circular-orbit method**	$GMm/r^2 = mv^2/r$; m cancels; $T = 2\pi r/v$
6	**Energy method for gravity**	$K + U$ conserved, $U = -GMm/r$; escape; satellite energies
7	**Choose the method**	clue → Kepler ratio, force, energy or vis-viva; cross-check

Extension: the shell theorem, vis-viva and Kepler 2 from angular momentum are in the supplementary notes.

Common mistakes: the five that cost most marks

1. The wrong distance

r is from the *centre*: $R + h$. Using h or R alone changes g, v and U.

2. Kepler's third law

$T^2 \propto a^3$ with the *semi*-major axis, so $T \propto a^{3/2}$; and same central body.

3. Where the Sun is, and where the planet is fastest

Sun at a *focus*; fastest at perihelion; $r_p = a(1 - e)$.

4. Mass in the orbit formulas

The satellite's mass cancels in v and T (and in v_{esc}); the *central* mass is what matters.

5. Signs of energy

$U = -GMm/r < 0$; $E < 0$ is bound;
$E = -GMm/2r$ (circle) or $-GMm/2a$ (ellipse).
A satellite that loses energy speeds up.

Lecture 7: momentum

Gravity as an external force; collisions and conservation laws use the same $K + U$ bookkeeping.

Lecture 8: rotation

Kepler's second law (equal areas) *is* conservation of angular momentum for a planet: $L = mvr$ constant gives $v_p r_p = v_a r_a$ (supplement now, derived in Lecture 8).

Lectures 10 and 11: thermal physics

The escape speed ($11.2\,\mathrm{km/s}$) decides which gases an atmosphere keeps: the speed distribution of molecules comes in Lecture 10.

Physical meaning

Later courses use the same inverse-square form $F \propto 1/r^2$ and $U \propto 1/r$ for the electric force; the method (sketch, centre, predict, force or energy) carries over unchanged.

Checklist: can I . . . CORE

- ☐ state Kepler's three laws and say what each is about?
- ☐ find r_p, r_a, a and e for an orbit?
- ☐ compare speeds at perihelion and aphelion?
- ☐ compare periods with $T_2/T_1 = (a_2/a_1)^{3/2}$?
- ☐ write the force between two masses, in vector form?
- ☐ add the gravitational forces from several bodies?
- ☐ find g above and below the surface, measuring r from the centre?
- ☐ tell weight from the gravitational force?
- ☐ find the surface gravity of a planet, and the mass of the Earth from g?
- ☐ find v and T for a circular orbit, and the mass of the central body?
- ☐ explain why the satellite's mass does not matter?
- ☐ use $U = -GMm/r$ with energy conservation?
- ☐ find the escape speed and the energy of a satellite?
- ☐ say what the sign of E tells me?
- ☐ choose between ratio, force and energy methods?
- ☐ use IDENTIFY → SET UP → EXECUTE → EVALUATE?

Next

Tutorial sheet 6: conceptual, guided, independent and application problems with answers, and an exam-style page of ten items. Supplementary notes: Kepler's second law from angular momentum, the shell theorem, $U = -GMm/r$, escape speed, $E = -GMm/2a$, vis-viva, and Kepler's third law for ellipses.

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